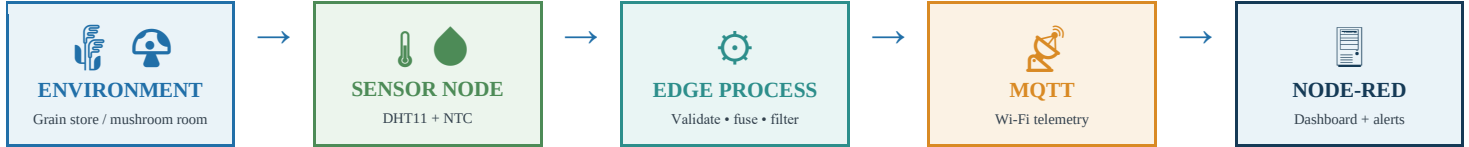


SMART GRAIN & MUSHROOM MONITORING SYSTEM

PROJECT PURPOSE

	Continuously monitor spatial variations in temperature, relative humidity at multiple locations; identify unfavorable conditions early; provide local + centralized warnings; and store telemetry for future analysis and validated predictive modelling.
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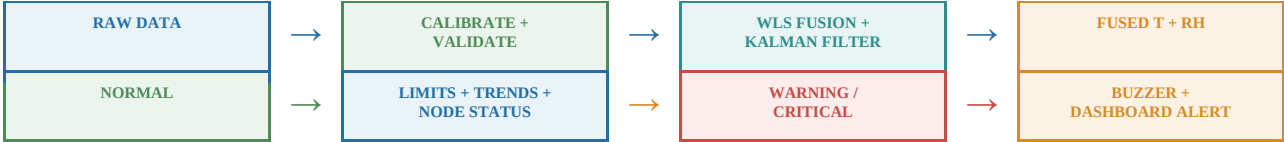
HOW THE SYSTEM WORKS



WHAT EACH SMART NODE DOES

 Temperature + Humidity DHT11 + NTC sensor fusion	 Connectivity Wi-Fi + MQTT node status	 Local alert Buzzer warning	 Location Temporary GPS initialization
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PROCESSING + EARLY-WARNING LOGIC



APPLICATION MODES & IMPORTANT ENGINEERING FACTS

 GRAIN STORAGE MODE - Tracks conditions associated with moisture-related deterioration. - Provides environmental-risk indication.	 MUSHROOM MODE - Checks whether temperature remain within selected cultivation ranges. - Uses stage/application-specific thresholds.	 SENSOR INTERPRETATION Thresholds must be documented from research and technical guidance.
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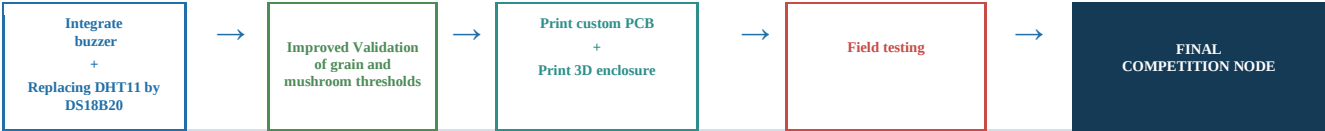
CURRENT PROGRESS & TECHNOLOGY STACK

 CURRENTLY WORKING / VALIDATED - Two operational wireless sensing nodes connected to a centralized Node-RED dashboard. - DS18B20 temperature + NTC temperature acquisition working. - Temperature calibration, validation, outlier rejection, WLS fusion and Kalman filtering architecture established. - Wi-Fi/MQTT live telemetry and multi-node dashboard visualization demonstrated. - OLED + NEO-6M GPS retained for prototype development/testing only.	 TECHNOLOGY STACK - Hardware: ESP32, DHT11, NTC, buzzer, temporary GPS, prototype OLED, power management. - Final hardware: custom PCB + enclosure; OLED removed; GPS used only when location initialization is required. - Embedded: calibration, validity checks, outlier rejection, WLS, Kalman filtering, condition logic, alarm control. - Communication/monitoring: Wi-Fi, MQTT, structured telemetry, Node-RED, trends, logging and alerts.
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VALIDATION, LIMITATIONS & TECHNICAL RISKS

 SYSTEM VERIFICATION & DEVELOPMENT STATUS - Completed: sensing, wireless telemetry, dashboard display and temperature-processing chain. - In progress: buzzer integration, Improving threshold validation, PCB/enclosure integration. - Required: long-duration field tests, representative grain/mushroom trials, network-loss recovery, power/battery characterization.	 LIMITATIONS / RISKS - Wi-Fi availability may interrupt live communication; robust reconnection is required. - Final thresholds, large-scale multi-node behavior and custom PCB performance still require validation.
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NEXT DEVELOPMENT MILESTONE



FUTURE PATH  Live monitoring → historical data → labelled dataset → analysis → ML training → validation → predictive monitoring